CONCEPT DRAWING SET

JBL Synthesis Theater, 24 × 18

Dolby Atmos 7.2.4 dedicated theater with two rows of seating, an acoustically transparent 120″ screen and an all JBL Synthesis loudspeaker and electronics package.

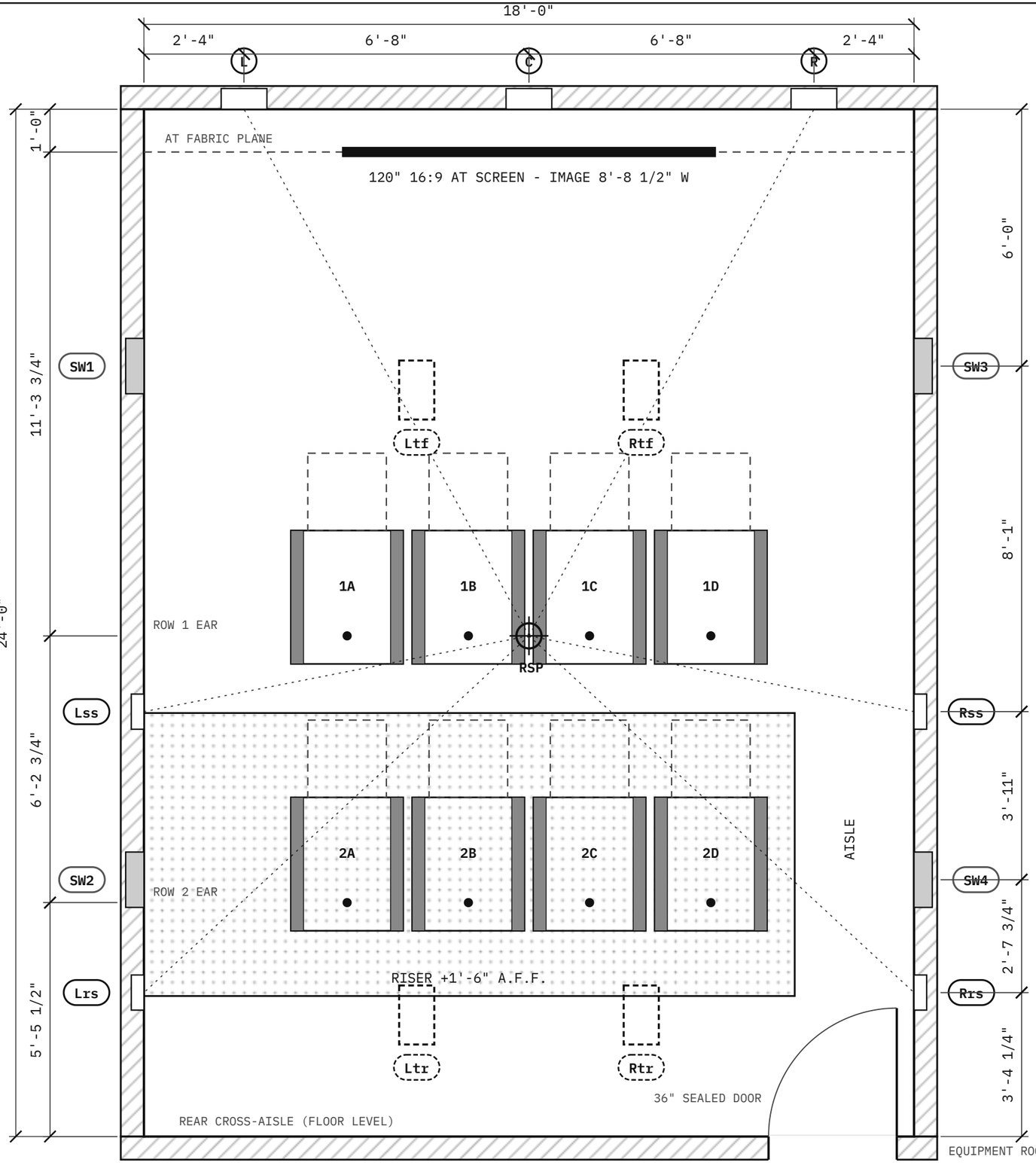
ROOM **24′ × 18′ × 9′ (110 m³)**	FORMAT **Dolby Atmos 7.2.4, 4 × SSW-3**	SEATING **2 rows × 4, riser 1'-6"**	SCREEN **120″ 16:9 AT, 8'-8 1/2" wide**
ROW 1 VIEW **42° horizontal**	PROCESSOR **SDP-58, 13/16 outputs**	EST. LEVEL **102+ dB peaks w/ SDA-7120**	BASS SPREAD **3.0 dB mean seat SD**

SHEET INDEX

T-001	Cover, sheet index, design basis
A-101	Floor plan and speaker layout
A-102	Reflected ceiling plan
A-201	Screen wall elevation
A-202	Side wall elevations
A-301	Longitudinal section
AV-501	Schedules
AV-601	Performance calculations and review

DESIGN BASIS AND ASSUMPTIONS

- Clean-room design using published manufacturer data and the CHT calculation catalog (Phase 3). Formula IDs on AV-601.
- Room 24′ L × 18′ W; ceiling 9′ is an assumption. Finished-face dimensions to be field-verified.
- Screen on the 18′ wall. Full-width acoustically transparent front wall; L/C/R in-wall behind it.
- Equipment rack outside the theater beyond the back wall; door on the back wall at the right aisle.
- Placement windows from Dolby home-theater guidance as commonly cited; verify against the current Dolby document.
- Output figures are estimates (±3 dB). JBL SCL-3 ceiling use and its grille size must be verified with JBL.



LEGEND

- In-wall loudspeaker (JBL SCL-2 / SCL-3)
- In-wall subwoofer (JBL SSW-3)
- Ceiling loudspeaker (above)
- Recliner, upright (footrest dashed)
- Riser platform
- Reference ray from RSP
- Reference seating position (RSP)

SPEAKER ANGLES FROM RSP

TAG	MODEL	AZ	EL	DIST	OK
L	SCL-2	28.4°	1.6°	14'-0"	3/8
C	SCL-2	0.0°	1.8°	12'-3 3/4"	-
R	SCL-2	28.4°	1.6°	14'-0"	3/8
Lss	SCL-3	101.1°	16.9°	9'-7"	4/8
Rss	SCL-3	101.1°	16.9°	9'-7"	4/8
Lrs	SCL-3	132.8°	12.8°	12'-7"	2/8
Rrs	SCL-3	132.8°	12.8°	12'-7"	2/8
Ltf	SCL-3	24.6°	41.3°	8'-5"	4/8
Rtf	SCL-3	24.6°	41.3°	8'-5"	4/8
Ltr	SCL-3	163.5°	31.0°	10'-9 1/4"	6/8
Rtr	SCL-3	163.5°	31.0°	10'-9 1/4"	6/8

PLAN NOTES

1. Dimensions to finished interior faces. Concept dimensions; field-verify before framing.
2. Front wall is a full-width acoustically transparent (AT) fabric plane; L/C/R are in-wall behind it.
3. In-wall speakers and subwoofers need 2x6 framing (SCL-2 >=152 mm cavity; SSW-3 5.1 in deep).
4. Speaker centres are shown; see AV-501 for X/Y/Z coordinates and AV-601 for seat compliance.
5. Ear points (dots) at 1.05 m above row floor; RSP is row-1 centre between seats 1B/1C.
6. OK = seats (of 8) inside the Dolby placement window for that channel.

CHT

CUSTOM HOME THEATER DESIGN

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
A (concept)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
3/8" = 1'-0" (on 11x17)

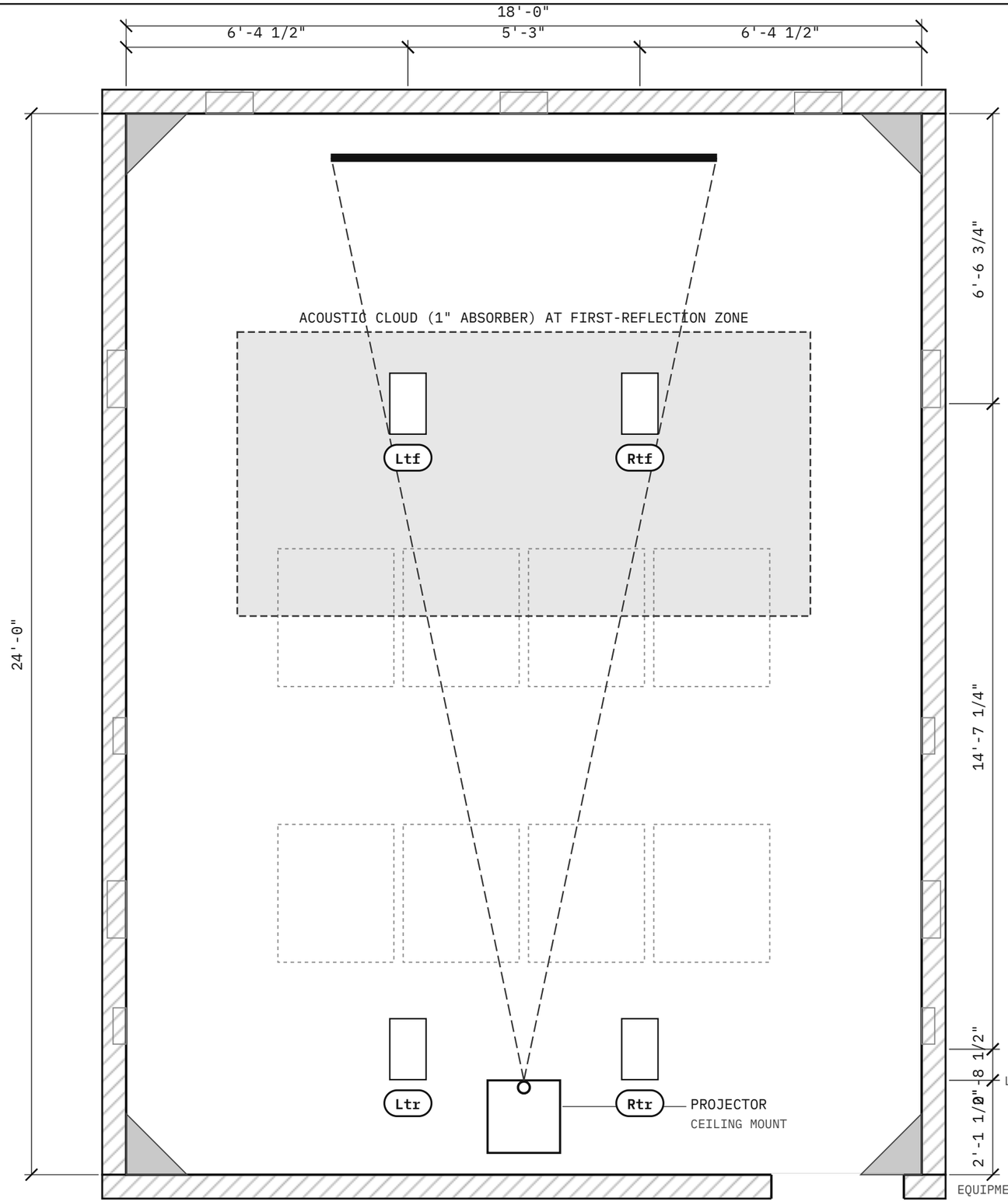
CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE

Floor Plan and Speaker Layout

A-101



LEGEND

- Ceiling loudspeaker (JBL SCL-3, verify ceiling use)
- Projector and mount (lens at front edge)
- Projected image edges
- Acoustic cloud / first-reflection zone
- Corner bass trap below (full height)
- Seats below (reference)

OVERHEAD SPEAKERS

TAG	X (L WALL)	Y (SCRN WALL)	EL@RSP
Ltf	6'-4 1/2"	6'-6 3/4"	41.3°
Rtf	11'-7 1/2"	6'-6 3/4"	41.3°
Ltr	6'-4 1/2"	21'-2"	31.0°
Rtr	11'-7 1/2"	21'-2"	31.0°

CEILING NOTES

1. Ceiling height 9'-0" (ASSUMED). Confirm joist direction and depth before locating ceiling speakers.
2. Projector lens at 20'-10 1/2" from screen plane, lens height 8'-1 1/2"; throw ratio 2.39:1.
3. Vertical lens shift required 75% of image height (published limit +/-80%; verify JVC definition).
4. Overhead speakers aimed toward midpoint between rows where the model allows rotation.
5. Provide a dedicated 20 A circuit and conduit (HDMI fibre + control) to the projector location.

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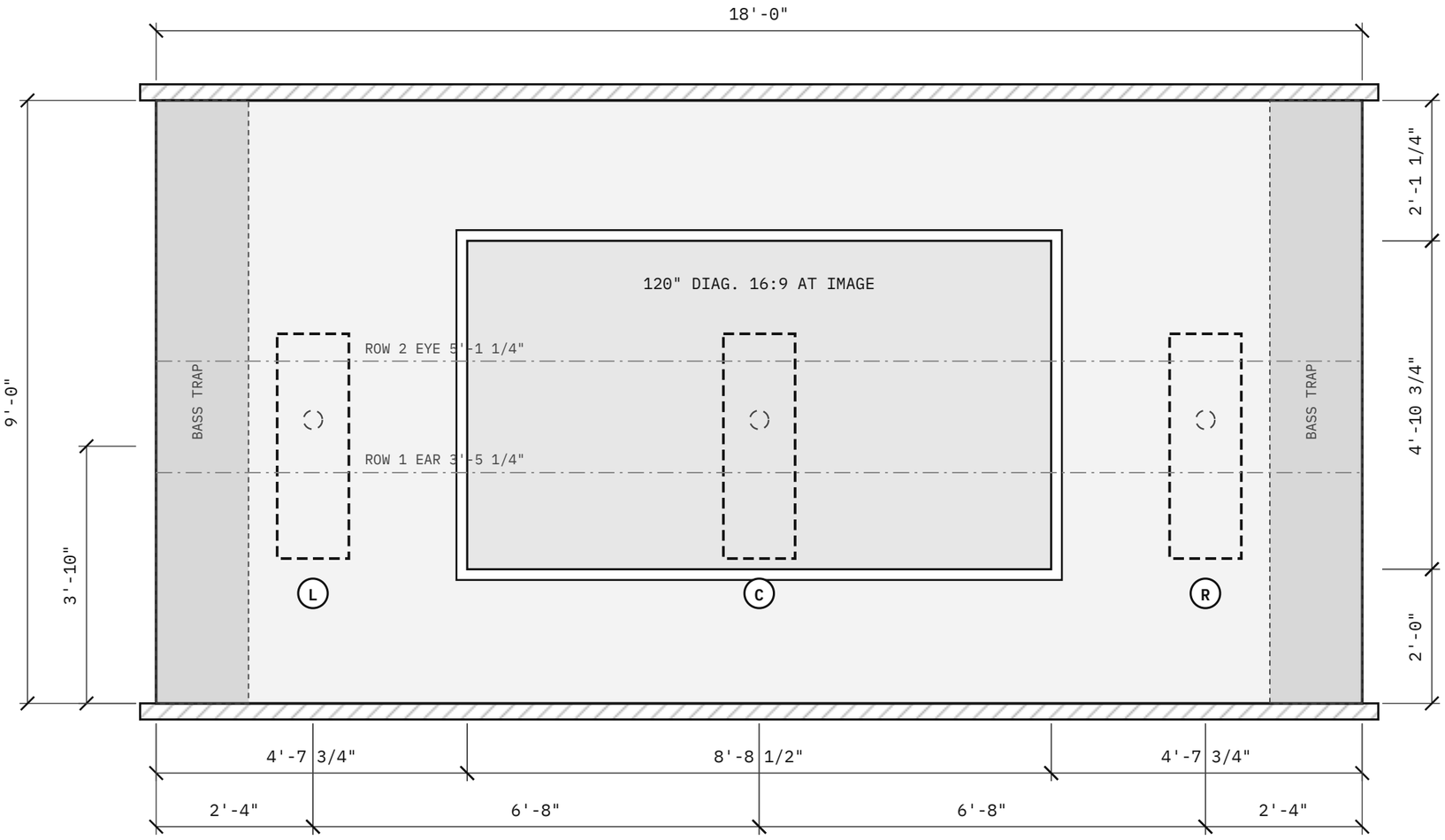
CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE

Reflected Ceiling Plan

A-102



ELEVATION NOTES

1. Image 8'-8 1/2" W x 4'-10 3/4" H; bottom of image 2'-0" A.F.F.; top 6'-10 3/4".
2. L/C/R JBL SCL-2 in-wall behind AT fabric, centred at 1.17 m (3'-10"); verify grille clear of screen frame.
3. Black, absorptive treatment behind the full AT plane; corner bass traps behind fabric.
4. Front-row vertical angle to top of image 16.3 deg (guidance ~15 deg, verify).

CHT

CUSTOM HOME THEATER DESIGN

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
A (concept)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
1/2" = 1'-0" (on 11x17)

CONCEPT DESIGN

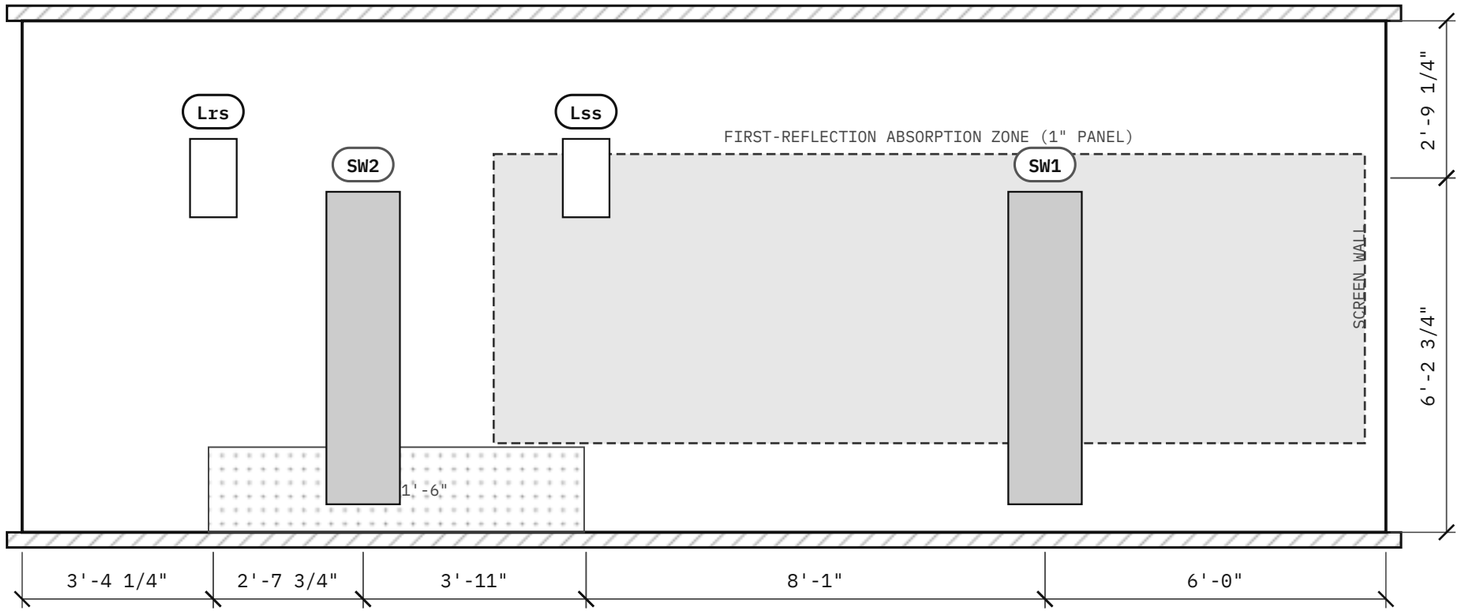
NOT FOR CONSTRUCTION

SHEET TITLE

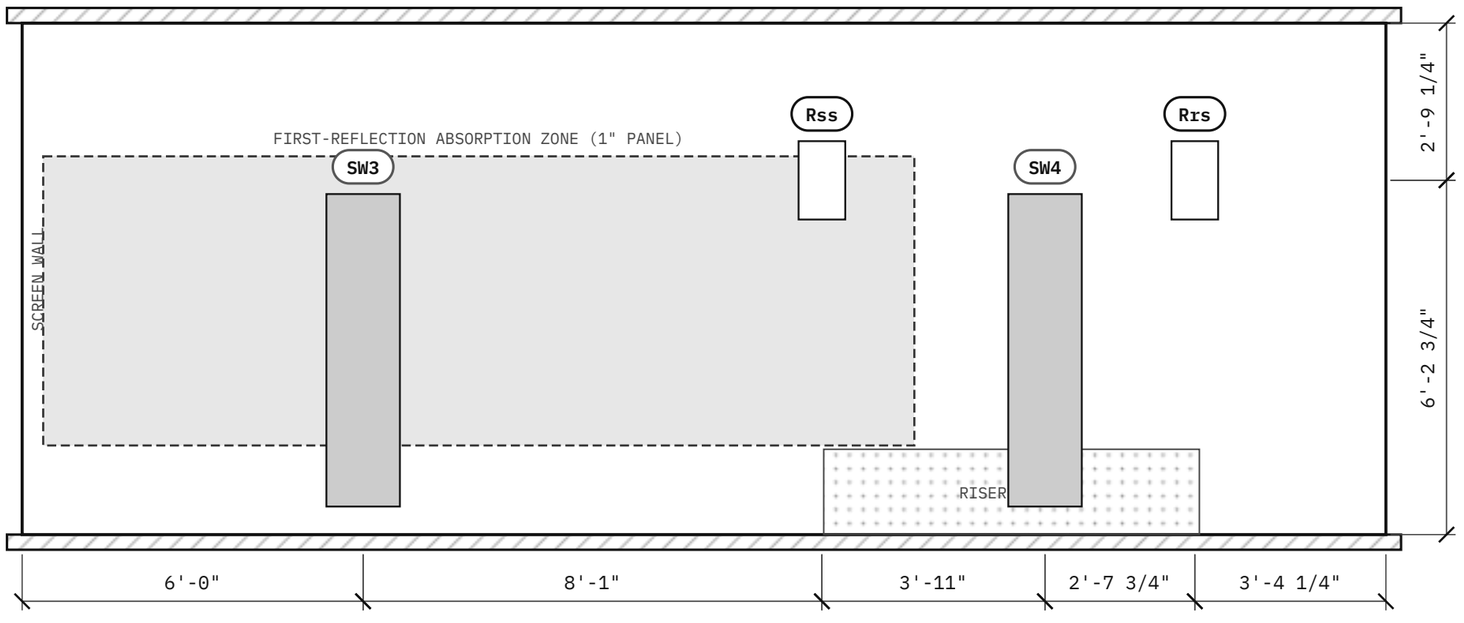
Screen Wall Elevation

A-201

LEFT WALL - VIEWED FROM INSIDE THE ROOM



RIGHT WALL - VIEWED FROM INSIDE THE ROOM



NOTES

1. Surrounds JBL SCL-3 in-wall at a common centre height to clear row-2 heads on the riser.
2. SSW-3 subwoofers are vertical in 2x6 walls, bottom of grille 6" A.F.F.; locate between studs per JBL template.
3. Absorption zones derived from first-reflection points of L/C/R for all eight seats, plus margin.
4. Treat the rear half of side walls with diffusion or leave reflective to avoid over-damping (see AV-601 RT60).
5. Subwoofer grilles may sit behind fabric; keep panels clear of sub cut-outs.

CHT

CUSTOM HOME THEATER DESIGN

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
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DATE
2026-11-09

DRAWN
CHT design engine v0.1

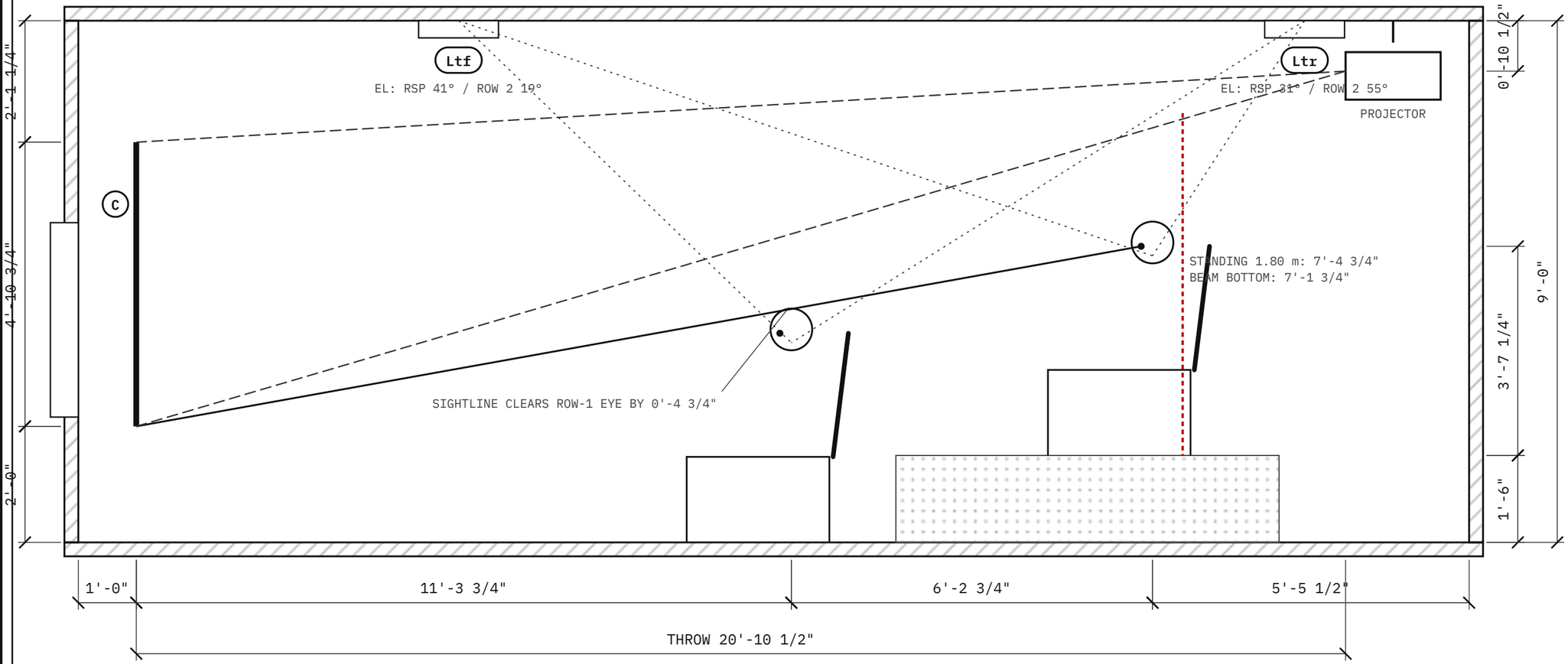
SCALE
3/8" = 1'-0" (on 11x17)

CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE
Side Wall Elevations

A-202



SECTION NOTES

1. Riser 1'-6" computed for a 120 mm sightline clearance (C-value) over row-1 eye to image bottom.
2. Viewing: row 1 41.6° horizontal / 16.3° to image top; row 2 27.7° / 5.8°.
3. Headroom on riser 7'-6". Overhead angles shown at RSP and row-2 centre (Dolby window 30-55°, verify).
4. Standing viewers on the riser intersect the lower image beam; consider a higher mount or rear-wall shelf (see AV-601).

CHT

CUSTOM HOME THEATER DESIGN

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
A (concept)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
1/2" = 1'-0" (on 11x17)

CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE

Longitudinal Section

A-301

Schedules: loudspeakers, equipment, amplification, cabling

LOUDSPEAKER SCHEDULE

TAG	MODEL	MOUNTING	X (FROM LEFT WALL)	Y (FROM SCREEN WALL)	Z (A.F.F.)	AZIMUTH @RSP	ELEVATION @RSP	DISTANCE	AIM
L	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	2'-4"	0'-0"	3'-10"	28.4°	1.6°	14'-0"	flush, on-axis
C	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	9'-0"	0'-0"	3'-10"	0.0°	1.8°	12'-3 3/4"	flush, on-axis
R	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	15'-8"	0'-0"	3'-10"	28.4°	1.6°	14'-0"	flush, on-axis
Lss	JBL Synthesis SCL-3	Left wall	0'-0"	14'-1"	6'-2 3/4"	101.1°	16.9°	9'-7"	flush, on-axis
Rss	JBL Synthesis SCL-3	Right wall	18'-0"	14'-1"	6'-2 3/4"	101.1°	16.9°	9'-7"	flush, on-axis
Lrs	JBL Synthesis SCL-3	Left wall	0'-0"	20'-7 3/4"	6'-2 3/4"	132.8°	12.8°	12'-7"	flush, on-axis
Rrs	JBL Synthesis SCL-3	Right wall	18'-0"	20'-7 3/4"	6'-2 3/4"	132.8°	12.8°	12'-7"	flush, on-axis
Ltf	JBL Synthesis SCL-3	Ceiling	6'-4 1/2"	6'-6 3/4"	9'-0"	24.6°	41.3°	8'-5"	yaw 163° / tilt 29°
Rtf	JBL Synthesis SCL-3	Ceiling	11'-7 1/2"	6'-6 3/4"	9'-0"	24.6°	41.3°	8'-5"	yaw -163° / tilt 29°
Ltr	JBL Synthesis SCL-3	Ceiling	6'-4 1/2"	21'-2"	9'-0"	163.5°	31.0°	10'-9 1/4"	yaw 25° / tilt 39°
Rtr	JBL Synthesis SCL-3	Ceiling	11'-7 1/2"	21'-2"	9'-0"	163.5°	31.0°	10'-9 1/4"	yaw -25° / tilt 39°
SW1	JBL Synthesis SSW-3	Left wall, vertical	0'-0"	6'-0"	3'-3"	-	-	-	-
SW2	JBL Synthesis SSW-3	Left wall, vertical	0'-0"	18'-0"	3'-3"	-	-	-	-
SW3	JBL Synthesis SSW-3	Right wall, vertical	18'-0"	6'-0"	3'-3"	-	-	-	-
SW4	JBL Synthesis SSW-3	Right wall, vertical	18'-0"	18'-0"	3'-3"	-	-	-	-

Coordinates: origin at front-left floor corner at the finished faces; X across the room, Y from the screen wall, Z above finished floor. Azimuth 0° = toward the screen.

EQUIPMENT SCHEDULE

QTY	ITEM	DESCRIPTION	USE
3	JBL Synthesis SCL-2	2.5-way dual 8in in-wall LCR	L / C / R
8	JBL Synthesis SCL-3	2-way dual 5.25in in-wall (surround / Atmos ceiling)	Surrounds + overhead (verify ceiling-mount approval)
4	JBL Synthesis SSW-3	dual 10in passive in-wall subwoofer	Subwoofers
1	JBL Synthesis SDP-58	16-channel immersive processor	Processor
2	JBL Synthesis SDA-7120	7-ch Class G amplifier	Full-range amplification (see SPL notes)
2	Subwoofer amplifier (TBD)	>= 600 W @ 4 ohm per channel, 2 ch	SSW-3 x 4
1	JVC DLA-NZ8 (example)	Laser D-ILA projector	Video
1	120in 16:9 AT screen (model TBD)	8'-8 1/2" x 4'-10 3/4" image, gain 1.0	Video
8	Theater recliner (model TBD)	864 mm pitch	Seating

Processor outputs used: 13 of 16 (11 full-range + 2 sub outputs; 4 subwoofers, front pair and rear pair).

AMPLIFIER CHANNEL MAP

CHANNEL	AMPLIFIER	OUTPUT
L	AMP-1 (SDA-7120)	1
C	AMP-1 (SDA-7120)	2
R	AMP-1 (SDA-7120)	3
Lss	AMP-1 (SDA-7120)	4
Rss	AMP-1 (SDA-7120)	5
Lrs	AMP-1 (SDA-7120)	6
Rrs	AMP-1 (SDA-7120)	7
Ltf	AMP-2 (SDA-7120)	1
Rtf	AMP-2 (SDA-7120)	2
Ltr	AMP-2 (SDA-7120)	3
Rtr	AMP-2 (SDA-7120)	4
SW1	AMP-S1 (sub amp, model TBD)	1
SW2	AMP-S1 (sub amp, model TBD)	2
SW3	AMP-S2 (sub amp, model TBD)	1
SW4	AMP-S2 (sub amp, model TBD)	2

SPEAKER CABLE SCHEDULE

RUN	EST. LENGTH	GAUGE	LOOP R	LOSS
L	18.3 m (60 ft)	12 AWG	0.190 Ω	-0.27 dB
C	16.2 m (53 ft)	12 AWG	0.169 Ω	-0.24 dB
R	14.2 m (47 ft)	14 AWG	0.235 Ω	-0.33 dB
Lss	14.0 m (46 ft)	14 AWG	0.231 Ω	-0.33 dB
Rss	8.5 m (28 ft)	14 AWG	0.140 Ω	-0.20 dB
Lrs	12.0 m (39 ft)	14 AWG	0.198 Ω	-0.28 dB
Rrs	6.5 m (21 ft)	14 AWG	0.107 Ω	-0.15 dB
Ltf	13.5 m (44 ft)	14 AWG	0.223 Ω	-0.32 dB
Rtf	11.9 m (39 ft)	14 AWG	0.197 Ω	-0.28 dB
Ltr	9.0 m (30 ft)	14 AWG	0.149 Ω	-0.21 dB
Rtr	7.4 m (24 ft)	14 AWG	0.123 Ω	-0.18 dB
SW1	17.3 m (57 ft)	10 AWG	0.114 Ω	-0.24 dB
SW2	13.7 m (45 ft)	12 AWG	0.142 Ω	-0.30 dB
SW3	11.8 m (39 ft)	12 AWG	0.123 Ω	-0.26 dB
SW4	8.2 m (27 ft)	14 AWG	0.136 Ω	-0.29 dB

Rack outside the back wall. Length = vertical drop + ceiling route (Manhattan) + 1.5 m drop + 2.0 m service loop. Gauge chosen so loop resistance ≤ 5% of 0.8 × nominal impedance (stand-in for minimum impedance).

Performance calculations and design review

SPEAKER PLACEMENT COMPLIANCE PER SEAT

CH	WINDOW	1A	1B	1C	1D	2A	2B	2C	2D
L	az 22-30°	11°	23°	33°	42°	7°	16°	24°	30°
R	az 22-30°	42°	33°	23°	11°	30°	24°	16°	7°
Lss	az 90-110°	110°	103°	100°	98°	47°	60°	67°	71°
Rss	az 90-110°	98°	100°	103°	110°	71°	67°	60°	47°
Lrs	az 135-150°	150°	138°	129°	122°	114°	105°	101°	99°
Rrs	az 135-150°	122°	129°	138°	150°	99°	101°	105°	114°
Ltf	el 30-55°	43°	43°	38°	32°	19°	19°	18°	16°
Rtf	el 30-55°	32°	38°	43°	43°	16°	18°	19°	19°
Ltr	el 30-55°	32°	32°	30°	26°	53°	55°	40°	29°
Rtr	el 30-55°	26°	30°	32°	32°	29°	40°	55°	53°

Placement found by search over wall and ceiling positions (Phase 3, A4). 7.2.4: 38/80 seat-channel pairs in window (48%); 7.2.6: 42/96 (44%).
Selected 7.2.4. Windows: Dolby home guidelines as commonly cited [verify].

OUTPUT AND AMPLIFICATION (ESTIMATE)

CH	MODEL	DIST (M)	W FOR 105+3 DB	SPK MAX, FREE-FIELD	SPK MAX, +ROOM	WITH SDA-7120, +ROOM
L	SCL-2	4.27	967	104.6	108.0	102.3
C	SCL-2	3.75	748	105.7	108.6	102.9
R	SCL-2	4.27	967	104.6	108.0	102.3
Lss	SCL-3	2.92	571	105.2	107.1	103.1
Rss	SCL-3	2.92	571	105.2	107.1	103.1
Lrs	SCL-3	3.83	983	102.8	105.8	101.8
Rrs	SCL-3	3.83	983	102.8	105.8	101.8
Ltf	SCL-3	2.56	439	106.3	107.9	103.9
Rtf	SCL-3	2.56	439	106.3	107.9	103.9
Ltr	SCL-3	3.29	722	104.2	106.5	102.5
Rtr	SCL-3	3.29	722	104.2	106.5	102.5

Target 105 dB peaks per channel + 3 dB margin at the RSP. "+room" adds the statistical reverberant field (Q = 5, R = 155 m²). Class E estimate, about ±3 dB.

SUBWOOFER CONFIGURATIONS (MODAL MODEL)

SUBWOOFER OPTION	FEASIBLE	MEAN SEAT SD (DB)	MAX SD (DB)
2x front wall, between C and L/R	yes	4.58	10.5
2x side-wall midpoints	yes	5.07	9.0
2x front corners	yes	5.24	16.0
4x front (between C-L/R) + back wall	yes	4.62	10.2
4x corners	**no** (conflict)	4.26	8.8
● 4x side walls at 1/4 and 3/4 length	yes	2.96	8.2
4x wall midpoints (front pair split around C)	yes	5.23	9.3

Rigid-wall modal sum, T60 = 0.4 s below 250 Hz, 8 seats, 20-100 Hz. Ranks options; absolute levels need measurement.

ROOM ACOUSTICS

BAND (HZ)	MODE @RSP FREE-FIELD (DB)	RT60 UNTREATED (S)	RT60 TREATED (S)
125	107.0	0.42	0.42
250	110.0	0.73	0.33
500	105.8	0.52	0.20
1000	113.2	0.45	0.17
2000	111.9	0.40	0.16
4000	114.4	0.37	0.15
	114.7		

Axial modes below 120 Hz: 23.4, 31.3, 46.9, 62.5, 62.5, 70.3, 93.8, 93.8, 117.2 Hz. Modes below 100 Hz: 25. Small-room reference ≈ 0.26 s. Coefficients are illustrative typical values; replace with tested ISO 354 / ASTM C423 product data.

VIDEO

SEAT	TO SCREEN (M)	HORIZ. ANGLE	TO TOP	TO BOTTOM	OFF-CENTRE
1A	3.44	37.8°	16.3°	-8.1°	20.6°
1B	3.44	41.6°	16.3°	-8.1°	7.1°
1C	3.44	41.6°	16.3°	-8.1°	7.1°
1D	3.44	37.8°	16.3°	-8.1°	20.6°
2A	5.35	26.5°	5.8°	-10.0°	13.6°
2B	5.35	27.7°	5.8°	-10.0°	4.6°
2C	5.35	27.7°	5.8°	-10.0°	4.6°
2D	5.35	26.5°	5.8°	-10.0°	13.6°

Image 8'-8 1/2" × 4'-10 3/4" (42.7 ft²). Projector throw 20'-10 1/2", ratio 2.39:1, lens shift 75%. Luminance: k=0.65, gain 1.0: 38.0 fL (130 cd/m²); k=0.45, gain 1.0: 26.3 fL (90 cd/m²); k=0.65, gain 0.85: 32.3 fL (111 cd/m²). 4K pixel structure resolves inside 2.38 m.

DESIGN REVIEW FLAGS

- Front-row vertical angle to image top 16.3 deg (>15 deg guidance [verify]).
- Standing 1.80 m person on riser reaches 2.26 m; beam bottom there 2.18 m -> casts shadow when standing.
- Subwoofers: est. free-field max 114.4 dB vs LFE target 115 dB; relies on low-frequency room gain (not modelled). Confirm with JBL CTA-2010 data.
- Treated mid-band RT60 0.18 s is well below the ~0.26 s small-room reference: risk of an over-damped room. Trade some absorption for diffusion (e.g. rear half of side zones).
- 125 Hz RT60 0.42 s vs mid-band 0.18 s: add low-frequency absorption (membrane/pressure traps) beyond the 4 corner traps.
- Axial mode coincidence near 62.5 Hz: [0, 0, 1] and [2, 0, 0] (L/W ratio).
- Axial mode coincidence near 93.8 Hz: [0, 4, 0] and [3, 0, 0] (L/W ratio).
- Axial mode coincidence near 125.0 Hz: [0, 0, 2] and [4, 0, 0] (L/W ratio).
- Level: SDA-7120 (~120 W @ 6 ohm, assumed) gives est. 101.8-103.9 dB peaks at the RSP: meets reference -5 dB, not full reference. Full 105 dB + 3 dB margin needs up to 983 W/ch (free-field basis) on L, C, R, Lss, Rss, Lrs, Rrs, Ltf, Rtf, Ltr, Rtr.
